UE24CS352A MACHINE LEARNING · MINI-PROJECT · PROBLEM 39

Lending Club P2P Loan Default Prediction

Reproducing a Stanford CS229 paper, and finding out why its results look too good

Aman Priyansh · PES1UG24AM030
Addanki Harsha Rajeswari · PES1UG24AM014

Team A-15 · Section A

THE PAPER

Can a model tell which P2P loans will be repaid?

On LendingClub, ordinary people lend to strangers, so the lender carries the risk.

1. Classification

Will the loan be **fully paid** or **default**?

Paper: **0.89** weighted precision and recall

2. Regression

What **yearly return** (NAR) will the loan give?

Paper: best R^2 **0.315** (Random Forest)

3. Strategy

Invest only in loans with a high predicted return.

Paper: **~1.7%** of loans, **~15%** a year

Li and Han, "LendingClub Loan Default and Profitability Prediction", Stanford CS229, 2018

REBUILDING THE BASELINE

What we were given, and what was missing

Item	What we found	Our fix
Dataset	LendingClub's download page no longer exists	Same data from Kaggle, filtered with the paper's rules
Cleaned data file	The code reads data_clean.csv, which is not in the zip	Rebuilt from the paper (2012–2015, finished loans)
Neural Network	No code file for the classifier	Rebuilt from the settings in the paper
2018 code	Breaks on today's libraries	Small updates; same maths
Random split	No fixed seed	Seed = 0, so anyone gets our numbers

Authors' code: 33 Python files, linked from the paper

REBUILDING THE BASELINE

The rebuilt data matches the paper closely

786,820

finished loans, 2012–2015

Paper: 745,529

18.6%

of loans defaulted

Paper: 19%

1,099

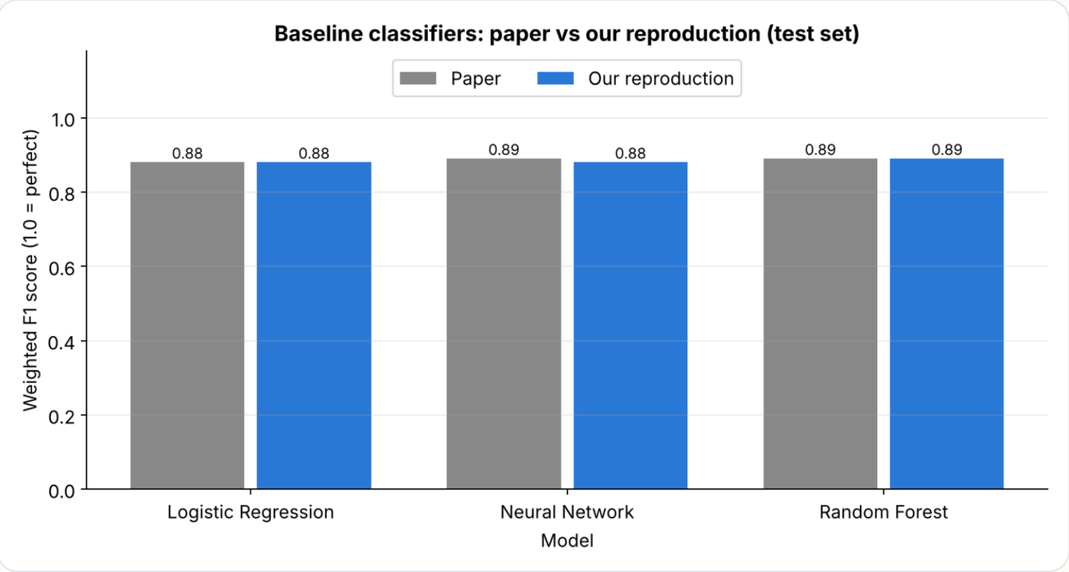
features after the authors' cleaning

Paper: 1,097

BASELINE RESULTS

We reproduced the paper

Test set	Paper	Ours
Weighted F1: Log. Reg. / NN / RF	0.88 / 0.89 / 0.89	0.88 / 0.88 / 0.89
R^2: Random Forest depth 10	0.315	0.281
Strategy: loans picked	~1.7%	1.34%
Strategy: yearly return	~15%	15.2%



Notebook 01_Baseline_Reproduction · Colab A100 · about 20 minutes

THE PROBLEM WE FOUND

Data leakage: the model sees the future

last_fico_range_high / low are the borrower's credit score at the **end** of the loan.

When someone stops paying, that score crashes, which almost gives away the answer. A real lender never has this number.

LIKE

predicting who fails an exam after seeing their final marks.

Also: a random split mixes old and new loans, there is no validation set or tuning, and gradient boosting was never tried.

OUR IMPROVEMENTS

Change one thing at a time (ablation)

Setting	Leaky columns	Train / test	Test loans
S1: paper setup	kept	random 70 / 30	235,929
S2: leak removed	removed	random 70 / 30	235,929
S3: honest test	removed	train 2012–14, test 2015	375,546

XGBoost on the GPU

Gradient boosting, with early stopping on a 15% validation set

Tuning

12 random settings, picked on validation ROC-AUC

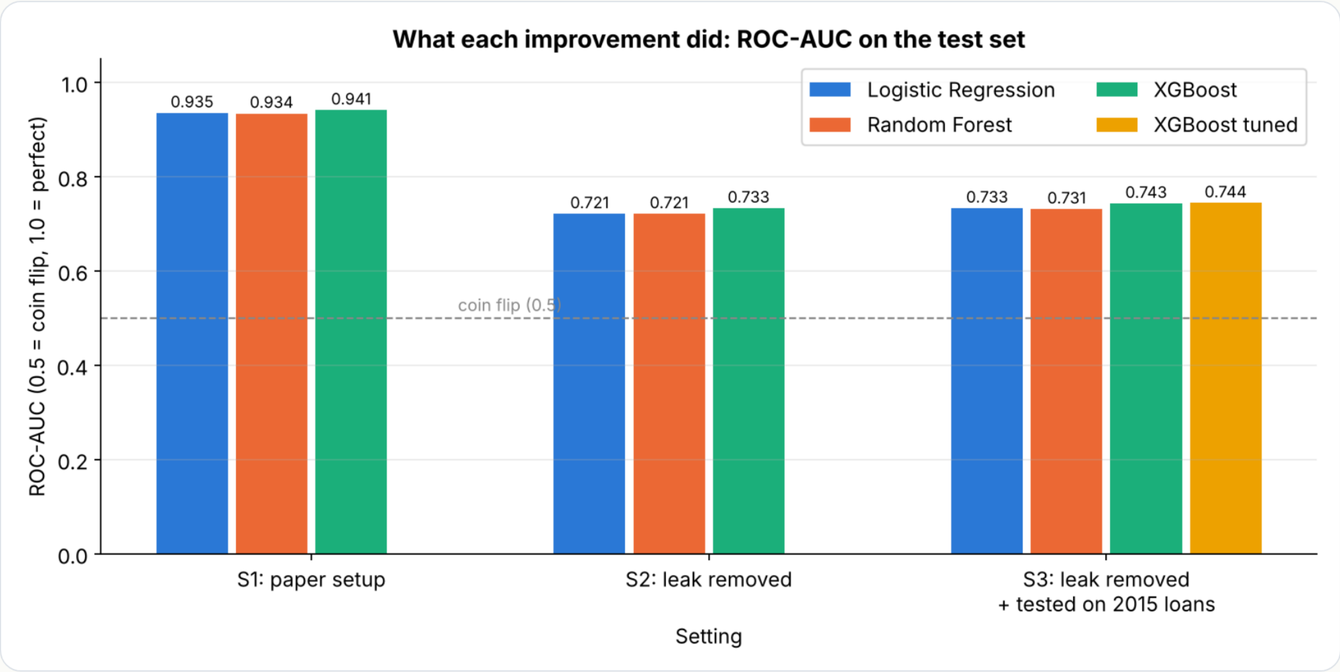
New metric

ROC-AUC: ranking quality over every cut-off

Notebook 02_Improvements · Colab A100 · about 15–20 minutes

CLASSIFICATION RESULTS

Removing the leak drops every model from 0.94 to 0.72



ROC-AUC on the test set · 0.5 = coin flip, 1.0 = perfect

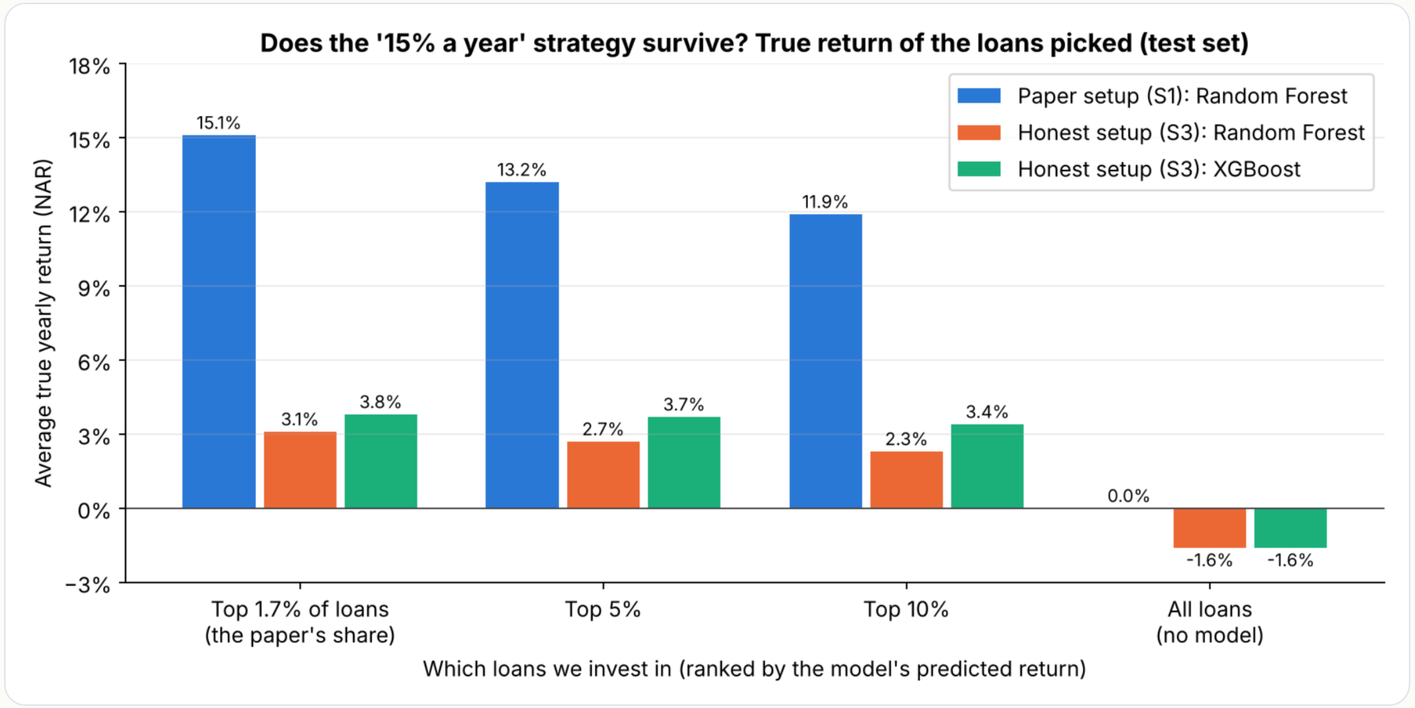
- The leak carries most of the paper's accuracy
- Testing on future (2015) loans does not hurt: 0.73–0.74
- XGBoost is best everywhere; tuned: **0.744**
- Tuning adds little (0.743 → 0.744)

INVESTMENT STRATEGY

The 15% return was mostly the leak

15.1% paper setup

3.8% honest setup



True yearly return (NAR) of the top 1.7% of test loans, ranked by predicted return

Still about 5 points better than investing in every loan (−1.6%).

CONCLUSIONS

What we learned

1. Reproduced

Same classifier scores (0.88–0.89) and the same ~15% strategy, after rebuilding the missing pieces

2. Leakage

The post-loan credit score carries most of the paper's accuracy

3. Honest skill

ROC-AUC 0.744 on future loans, with XGBoost best; tuning adds little

4. Honest return

About 3.8% a year, not 15%, but still better than investing blindly

LIVE DEMO

Pick a 2015 loan. See what the honest model predicts.

Tuned XGBoost, trained on 2012–2014 loans without the leaky columns. It shows the predicted chance of default next to what really happened.

Questions?

02_Improvements.ipynb · last cell (Gradio)